NANYANG TECHNOLOGICAL UNIVERSITY

SEMESTER 1 EXAMINATION 2018-2019

MH1812 - DISCRETE MATHEMATICS

December 2018 TIME ALLOWED: 2 HOURS

INSTRUCTIONS TO CANDIDATES

1. This examination paper contains **SEVEN (7)** questions and comprises **FOUR (4)** printed pages.

2. Answer **ALL** questions. The marks for each question are indicated at the end of each question.

3. Answer each question beginning on a **FRESH** page of the answer book.

4. This **IS NOT** an **OPEN BOOK** exam.

5. Candidates may use calculators. However, they should write down systematically the steps in the workings.

QUESTION 1.

(a) Prove that $\neg p \to \neg q$ and its inverse are not logically equivalent. **(10 marks)**

(b) Prove that $(q \wedge (p \to \neg q)) \to \neg p$ is a tautology using propositional equivalence and the laws of logic. **(10 marks)**

QUESTION 2.

Prove that $\sum_{j=n}^{2n-1}(2j+1) = 3n^2$ for all positive integers n. **(12 marks)**

QUESTION 3.

Find the solution to the recurrence relation $a_n = a_{n-1} + 2n + 1$ with $a_0 = 2$. **(10 marks)**

QUESTION 4.

(a) $x_1, x_2, \ldots, x_k$ are positive integers such that $\sum_{i=1}^{k} x_i = n$, for some positive integers k, n and $n \geq k$. How many distinct tuples of $(x_1, x_2, \ldots, x_k)$ are there?
(6 marks)

(b) How many distinct tuples of $(x_1, x_2, \ldots, x_k)$ are there for the above question if $x_1, x_2, \ldots, x_k$ are non-negative integers, rather than positive integers ?
(4 marks)

(c) How many bit strings contain exactly 5 '0's and 9 '1's if every '0' must be immediately followed by a '1' ? **(4 marks)**

QUESTION 5.

Prove by the method of membership table that

$$\overline{(A - B) \cup (B - A)} = (A \cap B) \cup (\overline{A} \cap \overline{B}).$$

(14 marks)

QUESTION 6.

Let $A = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$ and define the relation R as follows: $\forall x, y \in A$, $x\,R\,y$ *iff* $3^x \equiv 3^y \mod 5$.

(a) Prove R is an equivalence relation. **(8 marks)**

(b) List all the equivalence classes and all the elements in each class. **(8 marks)**

QUESTION 7.

Define a function $f : D \to \mathbb{Z}$ by $f(x) = x^2 + 5$, where $D = \{-4, -3, -2, -1, 0\}$.

(a) Find the range of the function. **(8 marks)**

(b) Find f^{-1}. **(6 marks)**

END OF PAPER